

2.5.1 Title, researchers and institution

קול קורא 2026/34 · משרד האנרגיה והתשתיות · יחידת המדען הראשי · מסלול אקדמיה

JouleFlex: A Model-Based Decision and Control Framework for Maximizing Useful AI Work under Dynamic Power Limits

JouleFlex: מודלים ובקרה חכמה למקסום תפוקת AI תחת מגבלות הספק משתנות

Principal Investigator (מנהל הפרויקט): Alexander Apartsin, Senior Lecturer, Department of Computer Science · **Institution:** Holon Institute of Technology (HIT)

2.5.2 Abstract / תקציר

English (299/300 words)

In July 2026 Israel's Electricity Authority froze new data-center grid-connection requests for 140 days: the queue had reached about 27,000 MW, roughly three times the country's average demand [2]. The binding limit is connection capacity, not generation. The PI's study [1] found that a GPU, the chip that runs AI, has an energy-optimal operating power, the Joule Point, well below its maximum; running at maximum, today's default and the level the grid is sized for, is the energy-worst choice. A capped GPU draws less power, so more GPUs fit behind the same connection. JouleFlex makes this a measured third option between full-power connection and refusal: connect on condition of flexibility.

Existing approaches limit power draw or switch GPUs off; this project asks instead how much useful AI work a given power and energy budget can deliver, a question that yields different operating strategies.

The PI's ELF dataset, the first dense enough to locate the Joule Point, holds about 5,500 measurements over 20 AI models and four GPU types; no public equivalent exists. Operating at the Joule Point uses 29 to 31 percent less energy per task at about 1.2 times the runtime; a fleet simulation on the measured curves delivers the same work with 18 to 45 percent less energy; and on the measured A100 curve a fixed connection delivers up to about 50 percent more AI work per megawatt at the chip, to be confirmed at the facility meter.

The project develops models and tools for operating and planning AI data centers under constrained and variable power. These include power-control and workload-allocation methods, pricing that rewards efficient use of flexible power, fleet planning for power-variable operation, and connection-scenario analyses showing how much of the connection queue firm-plus-flexible connections could serve by region, under technical, economic and policy constraints.

עברית (300/275 מילים)

ביולי 2026 הקפיאה רשות החשמל לתקופה של 140 יום את קבלת הבקשות החדשות לחיבור חוות שרתים לרשת החשמל, לאחר שתור הבקשות הגיע לכ-27,000 מגה-וואט, פי שלושה בקירוב מהביקוש הממוצע לחשמל בישראל [2]. האילוץ הכובל הוא קיבולת החיבור, לא כושר הייצור. מחקר קודם של החוקר הראשי [1] מצא כי למעבד GPU, השבב שמריץ בינה מלאכותית, קיים הספק עבודה אופטימלי מבחינה אנרגטית, ה-Joule Point, הנמוך בהרבה מההספק המרבי שלו; הפעלה בהספק המרבי, ברירת המחדל כיום והערך שלפיו מתוכננת רשת החשמל, היא דרך ההפעלה הגרועה ביותר מבחינה אנרגטית. מעבד מוגבל-הספק צורך פחות הספק, ולכן ניתן להציב יותר מעבדים מאחורי אותו חיבור לרשת. JouleFlex הופך זאת לאפשרות שלישית ומדודה, בין חיבור בהספק מלא לבין סירוב: חיבור בתנאי של גמישות.

הגישות הקיימות מגבילות את צריכת ההספק או מכבות מעבדים; פרויקט זה שואל במקום זאת כמה עבודת AI שימושית ניתן להפיק מתקציב הספק ואנרגיה נתון, שאלה המובילה לאסטרטגיות הפעלה שונות.

מערך הנתונים ELF, המדידה הראשונה הצפופה דיה לאיתור ה-Joule Point, כולל כ-5,500 מדידות על 20 מודלים של AI וארבעה סוגי GPU, ללא מקבילה ציבורית. עבודה בנקודה זו צורכת 29 עד 31 אחוז פחות אנרגיה למשימה, בזמן ריצה ארוך פי 1.2 בקירוב; סימולציית צי על העקומות המדודות מספקת עבודה זהה ב-18 עד 45 אחוז פחות אנרגיה; ועל עקומת ה-A100 המדודה, חיבור קבוע מספק עד כ-50 אחוז יותר עבודת AI לכל מגה-וואט ברמת השבב, ממצא שייבדק במונה המתקן.

הפרויקט יפתח מודלים וכלים להפעלה ולתכנון של חוות שרתים לבינה מלאכותית בתנאי הספק מוגבל ומשתנה: שיטות לבקרת הספק ולהקצאת עומסי חישוב, מנגנוני תמחור המתגמלים ניצול יעיל של הספק גמיש, תכנון צי GPU לעבודה בהספק משתנה, וניתוחי תרחישי חיבור המראים לרגולטור כמה מתור החיבור ניתן לספק תחת משטר חיבורים גמישים, לפי אזור, בהתחשב באילוצים טכניים, כלכליים ורגולטוריים.

2.5.3 Glossary

Term	Meaning
Power	How fast electricity is drawn at a given moment; watts (W) or megawatts (MW). The grid connection is sized by peak power.
Energy	The total electricity consumed: power $\times$ time; joules or kilowatt-hours. The bill and the carbon footprint are set by energy.
Power cap	A software limit on a GPU's power draw. Set with one command; takes effect in ~0.2 seconds. The control knob of this project.
TDP	A GPU's full rated power. Running at TDP is today's default.
Joule Point (JP)	The power cap at which energy per task is lowest for a given card; measured at 43–46% of TDP on large GPUs.
Operating point	The pair (GPU, power cap) a job runs at.
Power envelope	A limit on a facility's total power draw that can change over time, set by a budget, a price or carbon signal, or a grid instruction.
Digital twin / simulator	A simulator that replays the measured power-versus-rate curves of a described fleet to evaluate any policy or power envelope.
ELF	The PI's released dataset and measurement software: dense power-cap sweeps of

Term	Meaning
	20 AI models on 4 GPUs, ~5,500 measured rows.
SLO	Service level objective: the deadline or latency a job must meet.
Firm / flexible capacity	Grid connection capacity guaranteed at all times, versus capacity given on condition that the facility reduces power when instructed.
Noga	Israel's independent electricity system operator, which dispatches generation and would call on flexible loads.
Firm-plus-flexible connection	A grid connection combining a guaranteed (firm) part with a curtailable (flexible) part.
Energy-worst point	The operating point with the highest energy per task: full power (TDP).
Card / board	Card: one GPU. Board power is the GPU's own draw, from its built-in meter, excluding the host server, cooling and building losses.
Server input (BMC)	The power entering the server, read by its management controller (BMC); what the grid meters, before cooling and building losses.
Cap binds	The power cap actually limits the GPU (the GPU is busy enough to reach it).
LLM	Large language model: the AI model class behind chat and text services.
TAOZ	Israel's time-of-use electricity tariff.
FTE	Full-time-equivalent: one full-time researcher's work.
Bounding result	A documented limit on where the law or the JP holds; itself a deliverable.

2.5.4 Objectives of the program

2.5.4.1 The objective

The problem. In July 2026 the Electricity Authority stopped accepting new data-center connection requests for 140 days: the connection queue had reached about 27,000 MW, roughly three times Israel's average electricity consumption [2]. The binding limit is connection capacity at specific substations, not generation. **The Joule Point.** A GPU (the chip that runs AI) does its most work per joule below full power: lower its power cap (a software ceiling on its electricity draw) and output falls more slowly than power, so energy per task reaches a minimum, the **Joule Point (JP)**, before rising again. The PI's study [1] measured the JP across 20 models and 4 GPUs: well below the thermal design power (TDP, the card's full rated power), near-constant per card, and 29–31 percent less energy per task at about 1.2 times the runtime. A capped card draws less power, so more cards fit behind the same connection: on the measured A100 curve, up to about 50 percent more AI work per connected megawatt at the chip. Data centers still run cards at maximum power, the energy-worst point, and Israel's grid is sized for it.

The objective is to give the Ministry (policy) and the Electricity Authority (connection licensing) measured tools to license AI load on condition of flexibility, so that **each connected megawatt does more useful AI work**. The instrument is **JouleFlex**: models and tools for operating and planning AI data centers under constrained and variable power. Within 24 months the project extends the measured foundation to current hardware, builds the control engine and a simulator that replays the measurements, measures the flexibility a facility can offer, and analyzes the pricing and connection rules, with a stated minimum-success path (2.5.7.1); multi-site deployment follows the grant. Four capabilities, each built on the JP: **(C1) Power-control and workload-allocation methods**: the power-response law validated on current hardware and whole servers; job and transfer models that place the energy-optimal cap for any workload or unmeasured GPU; and a control engine that assigns jobs and caps together to hold a facility inside its power envelope while meeting deadlines. **(C2) Pricing rules evaluated in the simulator** that reward efficient use of flexible power instead of per-hour billing, which rewards the energy-worst point. **(C3) Fleet-planning methods for power-variable operation**: which cards, how many, with what power headroom, deliver the most work behind a variable connection. **(C4) Connection-scenario analysis for the regulator**: by transmission region, how much of the connection queue could be served under firm-plus-flexible connections at each flexibility level, under technical, economic and policy constraints. A simulator replaying the measured curves evaluates all four; C3 and C4 are scenario analyses run in the simulator on public Noga and Electricity Authority data (milestones in 2.5.7.1); they are studies, not deployed systems; everything is released openly with ELF-2, the extended dataset.

2.5.4.2 The research question

Can an AI data center run as a controllable electrical load under Israeli grid constraints: each GPU near its energy-optimal cap, the facility inside a power limit that changes over time, more AI work per unit of energy and of connected power, and every admitted job meeting its deadline?

Six sub-questions structure the work plan:

1. **Does the law generalize?** Do the power law and JP (measured on 20 models, 4 GPUs) hold on H100/L40S-class hardware, production inference software (running a trained AI model to answer requests), and shared GPUs?
2. **When is one cap per card enough,** and where must a job model replace the near-constant per-card JP?
3. **Where is the true optimum at the wall?** What is the JP at the server's electrical input, which the grid meters, not just the board?
4. **Can we skip the measurement?** Can an unmeasured GPU's JP be predicted from published specs plus at most one short test?
5. **Can allocation and power be optimized together?** Can job placement and per-card caps be chosen jointly under a time-varying power limit with every deadline met, and which card mix and power headroom deliver the most work behind a variable connection (C1, C3)?
6. **How much flexibility can a facility sell?** How much power can be shed, how fast, for how long, at what notice and cost; which pricing rules make flexibility profitable; and how much of the queue, by transmission region, firm-plus-flexible connections could serve (C2, C4)?

2.5.5 The proposed research

2.5.5.1 Scientific and technological background

The Israeli challenge is specific and pressing. Data centers, with AI the main growth driver, approach 945 TWh of electricity demand by 2030 [3]; Israel's grid is isolated, with no cross-border connections, so the 27,000 MW queue [2] makes connected megawatts, not energy, the binding national limit; today's operating decisions think in watts, not joules.

The discovery. The PI's study [1] (the ELF dataset) mapped how energy per task depends on the power cap: board power follows a simple empirical law in the computing rate θ ,

$$P(\theta) = P_0 + a \cdot \theta^6, \quad \text{median } R^2 = 0.99,$$

a fixed floor P_0 plus a superlinear term, so energy per task, $E = P/\theta$, is U-shaped in the cap with its minimum, the **JP**, at 43 to 46 percent of rated power on the large GPUs; the cap is one

software command, fast enough to follow grid signals [1]. These are GPU board numbers; server-input (BMC) power adds host overhead but excludes cooling and building losses (PUE), so WP1 and WP4 confirm the facility megawatt the grid sees.

2.5.5.2 Review of existing knowledge

Four research lines meet here. *Energy-proportional computing* [4] named the fixed, work-independent part of server power the central inefficiency; our floor P_0 measures it for GPUs. *Speed-scaling theory* proved that energy is minimized at a sub-maximal speed [5,6,7]; ELF gives the measured form for GPU inference. *GPU energy systems*, Zeus [8] and the ML.ENERGY benchmarks [9,10], search for a setting per job at runtime; the per-card-constant JP replaces the search with one measurement. *Production context*: cluster studies [11] and public traces [12] supply job arrivals; energy-aware GPU schedulers [13,14] leave device power at its default; methodology reviews [15] motivate the fixed protocol. Recent work power-caps inference servers at one recommended cap [20], finds capping ineffective for memory-bound decode [23], and runs AI data centers as grid assets by following a facility power target [21]; 2.5.5.6 states where JouleFlex departs from each.

2.5.5.3 Knowledge gaps

- **The law's reach and job dependence are unmapped.** The law is established for single jobs at fixed settings on three finely swept GPUs (A100, A10G, L4); its persistence on H100 and B200-class hardware, production serving software and shared cards is unknown; the per-card constant holds only when the GPU is busy enough that the cap binds, no model predicts a job's operating point from its features, and placing the JP on a new GPU still requires a full sweep [1].
- **All published numbers are board numbers.** The grid meters the facility; host overhead and power-supply losses shift the optimum upward by an amount WP1 measures; the chip-level megawatt figure awaits facility confirmation.
- **Allocation and power control are not jointly optimized.** The 18 to 45 percent fleet saving used a fixed budget, a fixed placement rule and caps chosen afterward; the joint problem, coupling discrete scheduling to continuous control through one power budget, is unsolved.
- **Flexibility is unquantified.** How much power an AI facility can shed, how fast, for how long, at what cost has never been measured or standardized; the Authority's hearing [19] is open and these numbers are not yet in it.
- **The incentive problem is identified but unsolved.** Per-hour billing makes the renter's cheapest choice the energy-worst point, and a per-joule surcharge is too weak [1]; which structural instruments work is open.

2.5.5.4 Description of the proposed research

JouleFlex builds one measured foundation and four tools on it: **WP1–WP2 measure and predict** the energy-optimal operating point per job and card; **WP3 controls** allocation and power in real time in a digital twin; **WP4–WP5 connect to the grid**, defining the flexibility a facility can offer and the rules that reward it.

2.5.5.5 Methods (work packages)

WP	Objective	Approach and output
WP1	Extend the energy-versus-power measurements to current GPUs, production software, and whole-server power.	Re-run the study's measurement sweep on current GPUs (H100/L40S class) and production serving engines (vLLM [16] and TensorRT-LLM, the software that runs AI models for users), reading power at three tiers: GPU board (NVML, the GPU's built-in power meter), host CPU (RAPL, the CPU's built-in meter), and server input, the wall tier (BMC, the server's management controller). The JP is where the tangent from the origin meets $P(\theta)$ ($dE/d\theta = 0$); a meter gain error leaves it unchanged and an additive offset shifts it, so JP location and per-card savings are gain-robust while absolute board-to-wall and megawatt figures carry the offset caveat. <i>Output</i> : the ELF-2 open dataset (board and server-input).
WP2	Predict any new card's or job's efficient operating point without a full measurement sweep.	A job model predicts a job's best power cap from its features; a transfer model, a parametric per-card scaling rule calibrated by one short test, places a new card's curve and is validated leave-one-out across the measured cards. Both report error bars and a range of validity; a short proof bounds the penalty of one cap per card. <i>Output</i> : the characterization kit (models and one-test protocol).
WP3	Decide, in real time, which job runs on which card and at what power cap, to do the most useful work within a changing power limit.	A controller that makes both choices together, since moving a job changes others' power budget, tested in a digital twin that replays measured curves against exact baselines. When power is tight it defers low-priority work and reports the cost. Per-card capping is already solved exactly [1]; the joint step is a heuristic scored against that bound. <i>Output</i> : the open control engine and digital twin.
WP4	Measure and characterize the flexibility a controlled facility can offer, in grid-contract form.	Measure the contractable quantities: a committed power envelope; a flexibility offer stated as MW shed of facility load measured at the connection point, ramp (seconds), hold duration (hours, through the evening peak), notice time, maximum events per season, recovery, work lost and cost; and an auditable delivery record. Demonstrate them on a rented metered server, no operator site required, and at fleet scale in the twin under an Israeli firm-plus-flexible scenario. WP4 measures both instruments defined in 2.5.6.2 (the connection coefficient and the two-tier curtailment product) at the connection point, against a stated baseline, with the work lost per MW shed. <i>Output</i> : the flexibility products and the Israel demonstration.
WP5	Evaluate, in the simulator, which pricing and connection rules reward efficient	An incentive model showing why per-hour billing rewards the energy-worst point and what changes fix it; a simulation-based valuation of the flexible-connection option from public figures

WP	Objective	Approach and output
	operation, and release everything openly.	(the TAOZ tariff, Noga marginal price, the Authority's capacity-holding payment and curtailment right [19]), reviewed by the advisor for energy-economics validity; each rule tested in the WP3 twin. <i>Output:</i> the policy report and the open release of the dataset, control engine, twin, and interfaces.

2.5.5.6 Novelty and originality

- **AI load as a grid resource, grounded in the measured optimum.** Unlike grid-following curtailment to a power target [21], JouleFlex runs each card at its measured energy-optimal cap and makes the resulting flexibility contractable and audited, quantified in MW, ramp, hold duration and recovery.
- **Energy first, power as the means.** Inference power-capping saves energy at a chosen cap [20] and can be ineffective for memory-bound decode [23]; JouleFlex sets power to the computed energy optimum per (card, job) where the cap binds, and every recommendation carries its capital cost in extra cards.
- **An open dataset with no equivalent.** WattGPU [22] and ML.ENERGY [9,10] report inference power at one operating point and open power-cap sweeps cover training or HPC; no sweep dense enough to locate the inference energy optimum existed before ELF.
- **The joint allocation-and-capping problem, to the measured optimum.** Placing jobs and setting every cap at the energy optimum is a joint problem grid-following controllers [21] do not address; the JouleFlex engine is a first solution on measured curves.

2.5.5.7 Fit to the Ministry's R&D goals

Submitted under topic 16.2 (AI and energy for data centers) of the Ministry's 2026 R&D policy [17], call 34/2026 [18]; the core deliverable answers 16.2 and the methods land on three 2026 priorities (8, 16.1, 6.3). Clause by clause (Chapter 1 numbering):

Clause	How the project addresses it
16.2 · טכנולוגיות תכנון וניהול הספקת אנרגיה לחוות שרתים	The core deliverable: the control engine manages a farm's energy supply in real time (each card at its optimum, the fleet held to a committed envelope); the twin sizes how many cards run behind a connection and how much power can be shed.
8 · התייעלות באנרגיה ובזרוע מערכות אנרגיה (priority 2026)	Energy efficiency of a new industrial load class (8.1, 8.3): 29–31 percent less energy per task at the card [1] and 18–45 percent per served job at the fleet (simulation), in software on installed hardware.
16.1 · שימוש בבינה מלאכותית ככלי עזר לפיתוח ומחקר בכל הנושאים לעיל (סעיפים 1-15) (priority 2026)	Machine learning as the instrument: the WP2 job model predicts a job's energy-optimal cap from its features, the scaling rule places a new card's curve from one short test, and the WP3 controller solves joint allocation-and-capping over the learned curves.
6.3 / 6.2 · ניהול אנרגיה מקומי	Demand management: cap actuation settles in about 0.2 s at the card and

Clause	How the project addresses it
וניהול ביקושים · בקרה וניהול רשת בחדירת מתחדשות גבוהה (priority 2026 : 6.3)	the demonstration targets a 60-second facility ramp; the project supplies the ramp, hold-duration and notice numbers the Authority's hearing [19] needs to define a curtailment product for AI load on an isolated grid.
15.3 / 15.1 · כלי מדיניות לייעול השימוש באנרגיה · מתודולוגיה וכלי עזר בקבלת החלטות	Policy tools for efficient energy use (15.1): WP5 tests billing and connection rules in the twin and shows which reward the energy-optimal point. Decision support (15.3): the regulator-facing report quantifies firm-plus-flexible connections against the queue.

2.5.5.8 Preliminary results

The preliminary results are the applicant's study *The Joule Point* [1] and its ELF dataset: about 5,500 instrumented measurements over 20 models and four GPUs (A100, A10G, L4 swept finely, T4 coarsely). Headline numbers: **29 to 31 percent** less energy per task at the JP (43 to 46 percent of TDP, runs 1.2 times longer); **18 to 45 percent** less energy per served job in a measured-curve fleet simulation at equal-or-better deadline compliance; cap actuation settling in **about 0.2 s** (191 ms measured). Supporting: law fit median $R^2 = 0.99$ per fixed-setting sweep (pooled per-card fits 0.88 to 0.92, Fig. A1); one static cap per card serves all twenty workloads at under one percent mean penalty; under strict deadlines capping saved 28 percent of energy while keeping 95 percent of workload value; under per-instance-hour billing the renter's cost optimum is the energy-worst operating point. The measurement software, the $P(\theta)$ fitting pipeline, the exact per-card capping solver and the fleet simulator behind these numbers are the PI's working code, and ELF is openly released, so every figure here is independently reproducible; WP1–WP3 extend this foundation.

2.5.6 Application: expected benefit and impact

2.5.6.1 What the project delivers, concretely

The project delivers five open artifacts: the ELF-2 dataset, the operating-point models with their one-test protocol, the control engine, the simulator built from the measurements, and the grid interfaces and policy report (Appendix Table A1: what each is, who uses it). They carry C1 (models, engine, simulator), C2 (policy report), C3 (simulator-based fleet planning) and C4 (queue scenario analysis by transmission region); all are released openly on delivery, crediting the Ministry's R&D program under call 34/2026 [18].

2.5.6.2 Benefit and impact

For the Israeli grid and regulator (the primary benefit). The freeze [2] affects four parties. The Ministry owns the policy and the freeze decision, with no measured basis for how much AI load can flex. The Electricity Authority (רשות החשמל), which licenses connections, must either grant connections the grid cannot back at specific substations, or keep the queue closed and block growth. Noga, the system operator, must hold the grid stable against a large, new and unmeasured load. The operators in the queue cannot build. None has the numbers to

license AI load on condition of flexibility: how much it can shed, how fast, for how long, at what notice and cost. JouleFlex produces exactly those numbers.

The third option rests on one fact: more work per joule is more work per connected megawatt, if the operator fills the same connection with more cards. A fixed-connection calculation on the measured A100 curve implies up to about 50 percent more AI work per megawatt at the GPU [1], to be confirmed at server input and facility level before any coefficient is recommended; the 27,000 MW queue includes speculative and duplicated applications, so any coefficient applies to approved capacity. WP5 puts a shekel-per-megawatt value on the flexible-connection option from public data, feeding the Authority's open hearing [19] while the rules are written. Connection constraints are locational, per substation and transmission region, so C4 is a scenario analysis by region of how much of the queue could be served under firm-plus-flexible connections at each flexibility level; with flexibility measured, committed and audited (WP4), the closed queue stops being all-or-nothing.

What the Authority could write into the 50 MVA connection track [19]. *Connection coefficient.* A facility committing to Joule-Point operation is licensed at k times the nameplate of its installed hardware, where k is its measured power at the Joule Point as a fraction of nameplate; WP1 measures k at server input and carries it to the facility meter with cooling and building losses (PUE), and the WP4 delivery record verifies it in operation. Worked example: a 100 MW applicant that commits to JP operation keeps a 100 MW connection and installs $100/k$ MW of nameplate; at the board value $k = 0.45$ [1] that is about 220 MW of hardware, delivering on the measured A100 curve up to about 50 percent more AI work than 100 MW uncapped. The facility k is higher, since host and cooling power do not fall with the cap. *Curtailment product in the system operator's direction.* A facility drops from its uncapped operating draw to the Joule Point on Noga's instruction, giving up roughly 40 percent of its operating power for about 20 percent of its speed, specified by MW shed against the measured uncapped draw (the MS0 memo fixes the exact per-card fraction), ramp (seconds), hold duration (hours, through the evening peak), notice time, maximum events per season, recovery, work lost and cost; the JP-and-below shed is a smaller second tier, each quantity measured by WP4 at the connection point and audited by the delivery record.

Training and human capital. The project builds research capacity at the AI-energy interface, which Israel lacks: a postdoctoral researcher trained across GPU energy measurement, ML-systems control and energy-market analysis; the open ELF-2 dataset, simulator and control engine as teaching material at HIT; and the Hebrew M4, M8 and M21 briefings to Ministry, Authority, Noga and operators.

For operators and planners. Operators (C1): one software command delivers 29 to 31 percent lower energy per task (measured) and 18 to 45 percent lower energy per served job (fleet simulation) on installed hardware; C2 pricing in place of per-hour billing lets a customer who accepts JP operation or curtailment windows pay less per unit of work. Planners (C3): the

simulator shows, before capital is committed, which fleet delivers the most work behind a variable connection.

2.5.7 Work plan

2.5.7.1 Stages, deliverables, duration and success metrics

Months	Stage	Deliverable	Success metric
M1–M6	WP1: existing measurement software extended; H100/L40S and vLLM sweeps; board and server-input instrumentation	D1: ELF-2 v1	Law fit $R^2 \geq 0.95$ per sweep (median 0.99 [1]); JP located, or its absence documented as a bounding result, per platform at board and server-input power
M3–M4	Preliminary regulator memo from the existing ELF data	MS0 (M4): memo (Hebrew)	Connection coefficient k and the full-power-to-JP shed fraction on the measured A100, A10G and L4 curves, board-level caveat stated, in the Authority's hands during the hearing [19] rule-writing window
M5–M8	WP1: shared GPUs, LLM phases, fine-tuning; sweeps where the constant should break	MS1: regime map; M8 brief (Hebrew)	Server-input (BMC) wall tier validated; validity domain documented, including the pre-registered JP shift (percentage points of TDP) at which the per-card constant breaks
M7–M12	WP2: job model; transfer model; one-test characterization protocol	MS2: models result	Job-model placement within 5 percentage points of TDP where the constant breaks, scaling-rule placement within 5 points, validated leave-one-out across the measured cards; protocol cost ≤ 8 hours and $\leq \approx 1,500$ per card model
M9–M16	WP3: twin; joint controller under fixed, price, carbon and curtailment envelopes	MS3: controller result	Energy per served job and SLO compliance versus uncapped, static-cap and cap-only baselines, with a pre-registered floor of at least 10 percent below cap-only at equal compliance (gain over an already-capped fleet); zero exceedance beyond the WP1 noise band in committed windows, transient violations under 1 percent of intervals
M14–M20	WP4: flexibility products; BMC-metered demonstration; Israel scenario	MS4: flexibility demonstration	Sheddable power in MW of facility load at the connection point and as a fraction of the measured uncapped draw: full-power-to-JP tier about 40 percent (exact fraction fixed by the MS0 memo); JP-and-below tier a pre-registered floor of at least 15 percent of the committed JP envelope within the ramp deadline at bounded SLO loss; measured on a BMC-metered server, projected to fleet scale in the twin; ramp within 60 seconds, hold duration in hours through the evening peak, notice time, recovery within 5 minutes, compatible with the curtailment right in

Months	Stage	Deliverable	Success metric
			the hearing [19]; delivery-record compliance and service cost; C3 check: a twin-recommended fleet beats a same-cost uncapped design by a pre-stated work margin
M17–M21	WP5: incentive model; connection valuation on published Israeli data; twin evaluation	MS5: policy report draft	Alignment condition per billing structure, confirmed in the twin; option value per MW and break-even curtailment frequency, with sensitivity to every price input; C4 check: a scenario analysis on at least 3 transmission regions built from public queue data, stating how much of each region's queue is served under firm-plus-flexible connections at each flexibility level
M21–M24	WP5: release, documentation, validation, publication	D2: open release; D3: final report	Public repositories with reproducing pipelines; metered validation; final scientific and policy reports

The M4 preliminary memo, the M8 technical brief and the M21 policy draft are named deliverables, in Hebrew, circutable to the Ministry, the Authority and Noga [19]; a bounding result is itself a deliverable. Minimum-success path: H100/L40S validation, one serving engine, BMC wall measurement, the job and transfer models, the joint controller, and one Israeli flexibility scenario; B200-class cards, a second engine, fine-tuning and shared GPUs are extensions as access allows.

2.5.7.2 Budget by stage

Stage	Share	Amount (₪)	Principal costs
M1–M8 (WP1; WP2 begins)	30%	148,314	Postdoctoral researcher; GPU rentals for cap sweeps
M9–M16 (WP2 ends; WP3; WP4 begins)	33%	163,145	Postdoctoral researcher; card-model rentals for transfer validation; twin and controller work
M17–M21 (WP4 ends; WP5)	25%	123,595	Postdoctoral researcher; BMC-metered rentals; flexibility campaign
M22–M24 (WP5: release and validation)	12%	59,326	Postdoctoral researcher; metered validation runs; documentation; dissemination and publication
Total		494,380	Full itemization in the מפרט תקציבי

2.5.7.3 Risks and mitigation

Risk	Impact	Mitigation
Production serving software (requests batched as they arrive, shifting phases) blurs the per-card constant	Static cap loses near-optimality for LLMs	Measured first (WP1, M1); the job model predicts the optimum where the constant breaks
Power-cap control or server-input (BMC) readout unavailable on some rented	Measurement campaign	AWS EC2 gives passthrough and root; too-high cap floors are exposed via the graphics clock (L4 [1]); named BMC-capable bare-metal vendors with dated

Risk	Impact	Mitigation
platforms	narrowed	quotes (GPU Mart and equivalents); fallback to an in-line metered PDU at the wall
Postdoc recruitment slips, or the single hire is lost mid-project	Scope narrows to the minimum path	PI runs the WP1 bench himself on his own measurement software; the minimum-success path is one-FTE; an M.Sc. student, funded from the unspent stipend, covers the gap; WP5 proceeds on public data

2.5.8 The applicant

Project manager and principal investigator: Alexander Apartsin, Senior Lecturer, Computer Science, HIT; single-PI, responsible for all five work packages, committing ~one day per week (~20 percent) over 24 months, no grant salary (§2.6).

Fit of expertise to the proposed work. The PI produced the measured foundation this project extends: study [1] delivered the ELF dataset, the power law and the Joule Point, and its measurement software, models and simulator are working code that WP1–WP3 extend directly. His engineering record is capacity-and-cost work of exactly this kind: system-sizing at Amdocs that contributed about \$10M in savings, and cost-aware AI systems in industry. He has led R&D teams in industry (Head of AI, Motorola Solutions Israel Innovation Center; Head of Data Science, Questar) and, as group leader, delivered the technical work and milestones of an Israel-U.S. BIRD binational grant and an Israel Innovation Authority Magneton grant. His academic thread is energy-efficient computing, from energy-efficient remote-sensing measurement [24] to the CalxNet study of efficient inference in deep models [25].

Research personnel. One postdoctoral researcher at 90% position, both years, extends the existing ELF measurement software, models and simulator under the PI's direct supervision, reported within a month of signature (2.5' נספח טז). Profile: GPU systems and energy instrumentation, ML systems, optimization or control; hiring-slip fallback in 2.5.7.3. WP4 and WP5, the flexibility products and the policy analysis on public Noga and Authority figures, run under the PI's direction with the advisor's guidance and are designed to be shared with the Authority and Noga during the hearing window [19].

Advisor. Prof. Moshe Tshuva (Afeka College of Engineering), a senior energy-systems engineer with two decades in energy management and efficiency, advises the project, unpaid, on the energy-systems and regulatory basis of the flexibility and valuation work.

Resources. The project owns no hardware and nothing waits on procurement: measurement and the WP4 demo run on rented AWS and BMC-metered servers at public list prices. HIT provides lab space, computing and administration.

2.5.9 Funding

No other, complementary, or parallel student funding is taken or sought (2.10, §1.5.2' נספח טז).

2.5.10 Bibliography

1. Apartsin, A. and Aperstein, Y. (2026). *The Joule Point: an Energy-Optimal Operating Point for AI Inference*. Preprint, September 2026. DOI: [10.13140/RG.2.2.20385.36964](https://doi.org/10.13140/RG.2.2.20385.36964) (CC BY 4.0), with the released ELF dataset.
2. JNS (22 July 2026). *Israel halts new data-center grid requests for 140 days amid surge in demand*. Press report on the Electricity Authority's temporary order: pending applications of approximately 27,000 MW, about three times Israel's average electricity consumption. <https://www.jns.org/news/israel-news/israel-halts-new-data-center-grid-requests-for-140-days-amid-surge-in-demand> (the Authority's order is published on gov.il).
3. International Energy Agency (2025). *Energy and AI: Energy Demand from AI*. <https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>
4. Barroso, L. A. and Hözl, U. (2007). *The Case for Energy-Proportional Computing*. IEEE Computer 40(12), pp. 33–37.
5. Yao, F., Demers, A., and Shenker, S. (1995). *A Scheduling Model for Reduced CPU Energy*. Proceedings of FOCS 1995, pp. 374–382.
6. Miyoshi, A., Lefurgy, C., Van Hensbergen, E., Rajamony, R., and Rajkumar, R. (2002). *Critical Power Slope: Understanding the Runtime Effects of Frequency Scaling*. Proceedings of ICS 2002, pp. 35–44.
7. Gandhi, A., Harchol-Balter, M., Das, R., and Lefurgy, C. (2009). *Optimal Power Allocation in Server Farms*. Proceedings of ACM SIGMETRICS 2009, pp. 157–168.
8. You, J., Chung, J.-W., and Chowdhury, M. (2022). *Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training*. arXiv:2208.06102. <https://arxiv.org/abs/2208.06102>
9. Chung, J.-W., Ma, J. J., Wu, R., et al. (2025). *The ML.ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization*. arXiv:2505.06371. <https://arxiv.org/abs/2505.06371>
10. Chung, J.-W., Wu, R., Ma, J. J., et al. (2026). *Where Do the Joules Go? Diagnosing Inference Energy Consumption*. arXiv:2601.22076. <https://arxiv.org/abs/2601.22076>
11. Hu, Q., Sun, P., Yan, S., Wen, Y., et al. (2021). *Characterization and Prediction of Deep Learning Workloads in Large-Scale GPU Datacenters*. arXiv:2109.01313. <https://arxiv.org/abs/2109.01313>
12. Alibaba Group (2020). *cluster-trace-gpu-v2020: production GPU cluster traces from Alibaba PAI*. <https://github.com/alibaba/clusterdata>
13. Lipe, E., Karia, N., Espenshade, C., Stein, C., Tantawi, A., and Tardieu, O. (2026). *Energy Efficient Scheduling of AI/ML Workloads on Multi-Instance GPUs with Dynamic Repartitioning*. arXiv:2606.25082. <https://arxiv.org/abs/2606.25082>

14. Lu, T. and Reda, S. (2026). *Agentic CPU-GPU Scheduling for Heterogeneous AI Workloads*. arXiv: 2607.22242. <https://arxiv.org/abs/2607.22242>
15. Rodriguez, C., Degioanni, L., Kamení, L., Vidal, R., et al. (2024). *Evaluating the Energy Consumption of Machine Learning: Systematic Literature Review and Experiments*. arXiv: 2408.15128. <https://arxiv.org/abs/2408.15128>
16. Kwon, W., Li, Z., Zhuang, S., et al. (2023). *Efficient Memory Management for Large Language Model Serving with PagedAttention*. Proceedings of SOSP 2023. arXiv: 2309.06180. <https://arxiv.org/abs/2309.06180>
17. Ministry of Energy and Infrastructure (2026). *R&D Support Policy for 2026* (in Hebrew). Chapter 1 clauses 6, 8, 15, 16; Chapter 2 priority list.
18. Ministry of Energy and Infrastructure (2026). *Call for Proposals 34/2026 for Research Funding in Energy and Earth and Marine Sciences* (in Hebrew).
19. Calcalist (June 2026). Report on the Electricity Authority's public hearing: a dedicated connection track for transmission-grid consumers of 50 MVA and above, including a system-operator right to curtail consumption at peak and payments for holding connection capacity. Under public consultation; cited as regulatory direction, not settled regulation. <https://www.calcalist.co.il/calcalistech/article/rylkxexmgx> (in Hebrew; the hearing documents are published on gov.il).
20. Ma, Y., Srinivasan, S., and Wang, X. (2025). *Power Capping of GPU Servers for Machine Learning Inference Optimization*. Proceedings of ICPP 2025 (54th International Conference on Parallel Processing). DOI: [10.1145/3754598.3754670](https://doi.org/10.1145/3754598.3754670).
21. Colangelo, P., Coskun, A. K., Megrue, J., et al. (2025). *AI data centres as grid-interactive assets*. Nature Energy. DOI: [10.1038/s41560-025-01927-1](https://doi.org/10.1038/s41560-025-01927-1); field demonstration in Phoenix, AZ, arXiv: 2507.00909.
22. Fadel Argerich, M., Fürst, J., and Patiño-Martínez, M. (2026). *WattGPU: Predicting Inference Power and Latency on Unseen GPUs and LLMs*. arXiv: 2607.02391. <https://arxiv.org/abs/2607.02391>
23. Ma, B., Afzal, A., Eitzinger, J., and Wellein, G. (2026). *The Illusion of Power Capping in LLM Decode: A Phase-Aware Energy Characterisation*. arXiv: 2605.11999. <https://arxiv.org/abs/2605.11999>
24. Apartsin, A., Cooper, L. N., and Intrator, N. (2014). *Energy-Efficient Time-of-Flight Estimation in the Presence of Outliers*. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 7(4). DOI: [10.1109/JSTARS.2013.2295324](https://doi.org/10.1109/JSTARS.2013.2295324).
25. Aperstein, Y. and Apartsin, A. (2026). *CalexNet: Soft Cascade-Aligned Training and Calibration for Lightweight Early-Exit Branches*. Electronics 15. DOI: [10.3390/electronics15102149](https://doi.org/10.3390/electronics15102149).

Appendix A: Figures (within the 5-page allowance of 2.2§ 1'נספח א')

JouleFlex · 34/2026 קורא א' · Appendix to the research proposal, permitted under §2.2 (graphs, tables and diagrams, up to 5 A4 pages, outside the 10-page limit). All figures are measured results from the applicant's study [1].

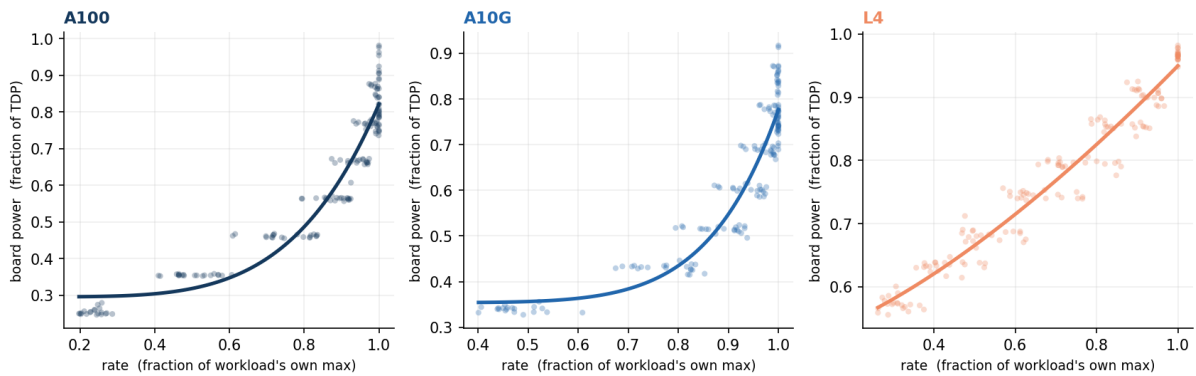


Figure A1. The response law is a card property. One panel per GPU (A100, A10G, L4); each dot is a measured operating point for one of the 20 workloads in the loaded regime, rate normalized to the workload's own maximum and power to the card's TDP. All workloads follow a single fitted law per card, $P(r) = P_0 + a \cdot r^b$ (pooled per-card R^2 of 0.88 to 0.92; per-sweep fits higher, median 0.99), showing the exponent travels with the card rather than the model. This is the empirical basis for the one-time per-card characterization that WP2 turns into a transfer model.

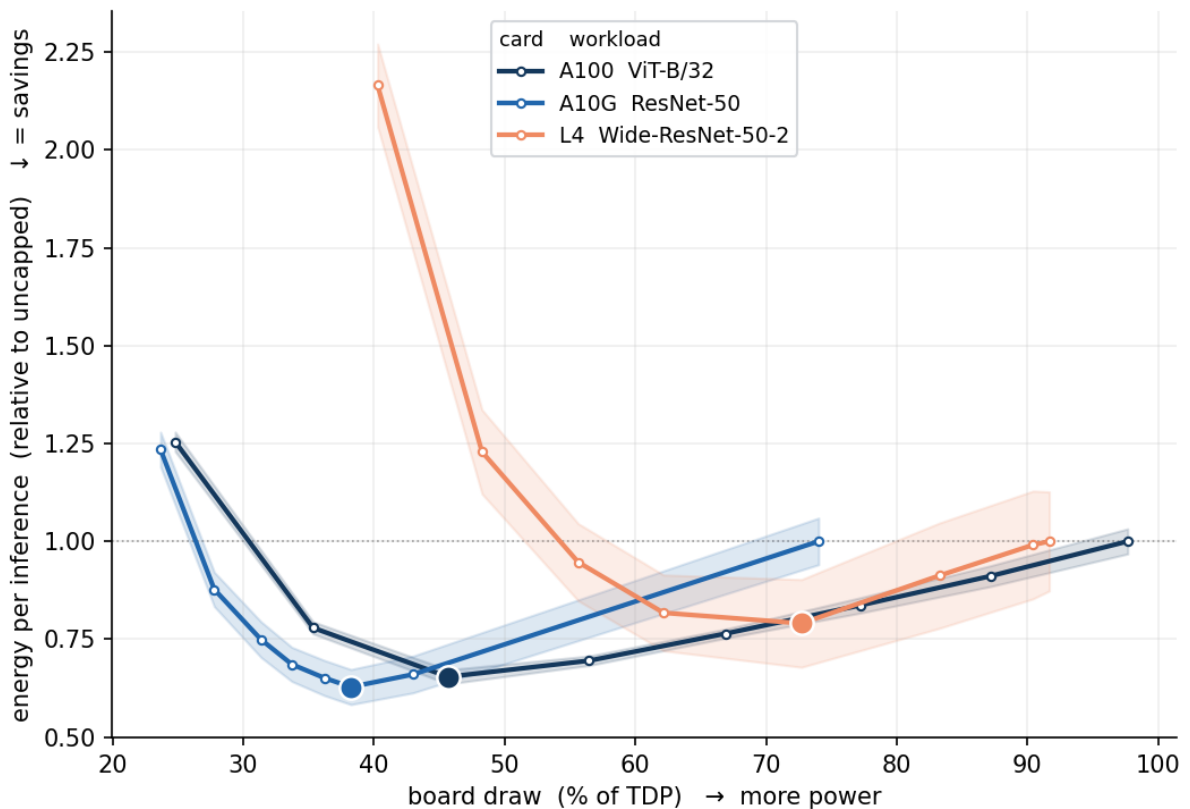


Figure A2. Every card has an energy U-shape. Energy per inference, normalized to each card's

own uncapped value, against board draw as a fraction of TDP; the dot marks the JP, the shaded band one standard deviation over repetitions. On the large GPUs the minimum sits at 43–46 percent of TDP and capping to it cuts energy per inference by 29–31 percent; on the L4 the interior minimum lies below the power cap's reachable floor and is exposed by the graphics clock instead, mapping which actuator applies to which card class.

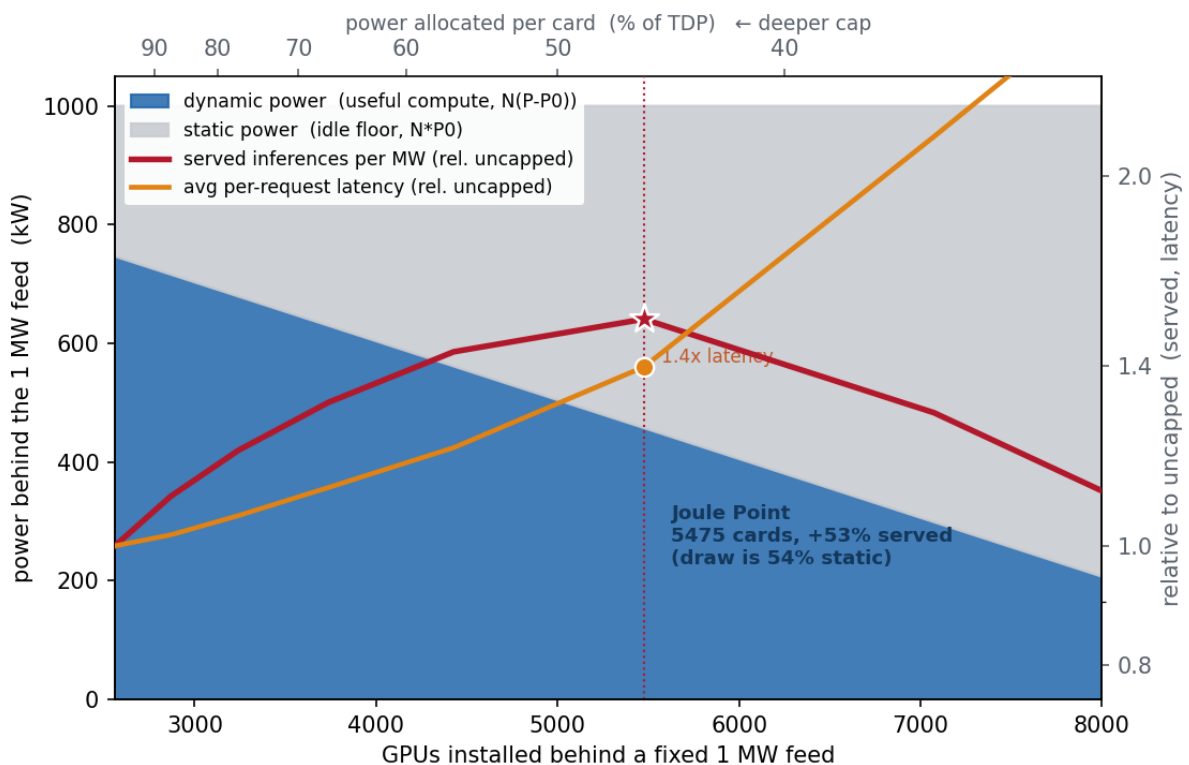


Figure A3. When power, not capital, binds, the JP is the interior optimum. Behind a fixed 1 MW feed (A100, ViT-B/32 at batch 64), installing more GPUs forces each to a deeper cap, shifting the megawatt from useful dynamic compute (blue) to the static idle floor (grey). Served inferences per megawatt (red) peak at the JP, near 5,500 installed cards, about 50 percent above the uncapped configuration. This is the capacity lever the Israel demonstration (WP4) operationalizes: under a rationed connection, efficiency becomes throughput. The same measured curves, allocated exactly across a fleet, dominate uncapped operation at every power budget from 100 down to 40 percent of total TDP; the preliminary measured-curve fleet simulation result appears in section 2.5.5.8.

Table A1. The five artifacts: what each is, who uses it

The project delivers five named artifacts, each with a defined user and use:

Artifact	What it is	Who uses it, for what
ELF-2	An open dataset: power, throughput and latency at every operating point, board to wall, on current GPUs and production serving software	Researchers and vendors, as ground truth; every other deliverable is built and validated on it

Artifact	What it is	Who uses it, for what
The characterization kit	Software plus protocol: a one-test procedure and the two fitted models placing the JP of a job or of a new, unmeasured card within the validated hardware domain, in hours	A GPU engineer or operator qualifying a new card model or a new serving stack before deployment, at known cost
The control engine	An open-source control engine that assigns jobs and sets caps to keep a fleet inside its envelope, evaluated in the twin and demonstrated on a metered rented server before any deployment	An AI data-center manager, to cut energy per job and honor a committed envelope; its report doubles as the compliance record
The digital twin	A simulator replaying the measured curves of a described fleet and job mix, evaluating any policy or envelope scenario exactly before hardware is touched	An operator testing policies before deployment; a planner or regulator asking what a proposed facility could deliver and shed under a given connection
The grid interface and policy report	A blueprint: open formats for the envelope commitment, the flexibility offer and the delivery record, plus a regulator-facing report quantifying firm-plus-flexible connections against the national queue	The Electricity Authority and Noga, as the technical basis for offering conditional connections and auditing compliance

Table A1. The chain of use is direct: ELF-2 calibrates the models, the models feed the controller, the twin proves the controller before deployment, and the grid interface turns the controlled facility into something the regulator can contract with.